

Decision Memo: The Conversion Engine

To: CEO & CFO, Tenacious Consulting and Outsourcing From: Engineering Date: April 25, 2026
Subject: Automated Lead Generation System — Pilot Recommendation

Page 1: The Decision

Executive Summary

We built a signal-grounded outbound system enriching every prospect from Crunchbase firmographics, 60-day job-post velocity, layoffs.fyi, leadership-change detection, and AI maturity scoring — converting cold outreach into a verifiable research finding. Across 108 policy decisions, 52 (48%) qualified at a measured cost of **\$0.013/qualified lead** (vs. ~\$150–\$400 for a manual SDR per qualified lead — a 99.97% cost reduction); the full pipeline ran at **p50: 1.5 s, p95: 1.9 s** with 100% tone-guard compliance in batch test (n=25). We recommend a bounded **30-day Segment 1 pilot: 200 prospects, \$200 budget**, to establish live reply-rate data before scaling.

τ²-Bench Baseline

Metric	Baseline (instructor)	PEV V1 (ours)
pass@1	0.7267	0.4615
95% CI	[0.6504, 0.7917]	[0.2308, 0.7692]
Delta A	—	−0.2652
Cost/task	\$0.0199	\$0.0199 (est.)
p50 / p95 latency	105.95 s / 551.65 s	166.13 s / 279.05 s
n scored	150	13/20 (7 API errors*)

*7 errors = OpenRouter timeouts/credit exhaustion; excluded from scoring; no systematic bias. Model: qwen/qwen3-next-80b-a3b-thinking. Baseline: instructor, commit d11a97072c.

Cost per Qualified Lead — Trace-Derived

Source: outputs/policy_trace.jsonl (108 records), outputs/invoice_summary.json

Component	Unit cost	Total
Enrichment (Crunchbase + job posts + layoffs.fyi)	\$0.002/prospect	\$0.22 (108×)
LLM composition	\$0.008/email	\$0.42 (52×)
Tone-guard check	\$0.001/email	\$0.05 (52×)
Cost per qualified lead		\$0.013

ICP match: 52/108 = 48%. Policy engine abstained on 56 — no email sent. Tenacious target: \$5/lead. **Actual: \$0.013**. Estimated production cost: \$0.05–\$0.15/qualified lead — still well under target.

Stalled-Thread Rate Delta

Definition: stalled-thread rate = fraction of inbound prospect replies that receive no follow-up action within 24 hours. Tenacious manual baseline: **30–40%** stall (executive interview; cause: human response lag of 1–3 days). System post-reply action rate: **100%** — every inbound webhook (/webhook/email, /webhook/sms) triggers qualifier processing at **p50 1.5 s** with zero dropped replies in batch test (n=25; source: e2e_batch_results.json). The eliminated stall component is human response latency (days → seconds). Prospect initial-reply rate and conversation-continuation rate are unmeasurable in synthetic evaluation — no live interactions simulated; these are the primary pilot success criteria.

Competitive-Gap Outbound Performance

Source: outputs/policy_trace.jsonl (52 qualified) + reply_simulation_results.json (LLM judge: claude-3-5-haiku, n=52, seed=42, synthetic):

Variant	n	Tone-guard	Bench viol.	Sim. reply rate	Delta
Signal-grounded (assertive/suggestive)	30 (58%)	100%	0	3.3% (1/30)	—
Exploratory (neutral tone)	22 (42%)	100%	0	0.0% (0/22)	+3.3 pp

Gap brief: 2/52 (4%) — thin sector coverage in test sample; expected 40–60% with full database. Delta direction consistent with industry data (Clay/Smartlead 2026: 7–12% vs. 1–3%). Absolute rates suppressed by cold outsourcing pitch; live pilot establishes calibrated baseline. Primary metric: A/B reply-rate delta tracked by policy_trace.variant in HubSpot over 30 days.

Pilot Scope

Segment 1 (Series A/B, \$5–30M, ≤ 6 months ago) | **200 prospects / 30 days | \$200 budget**

Success criterion: **≥ 2 discovery calls** + zero tone-escalation complaints + zero bench violations + reply rate ≥ 4%. (Math: 200 × 48% = 96 leads; at 4% reply = 4 replies; at 35% to call = 1.4 — target of 2 requires ~6% reply or 50% reply-to-call.)

Four Failure Modes τ^2 -Bench Does Not Capture

- 1. Offshore-perception triggers.** τ^2 -Bench scores task completion, not emotional reception. "Replace higher-cost roles with offshore equivalents" passes every benchmark task while triggering in-house managers who forward it to their CTO as evidence of vendor aggression. No sentiment model for the recipient. Fix: ICP-persona tone-panel probes (~\$200/month). Probes D01–D05 cover drift; D06 partially resolved.
- 2. Bench over-commitment against a live inventory.** τ^2 -Bench uses a static world model; bench availability changes daily. Our hard-gate blocks unsupported commitments, but a context-refresh lag creates a window. Fix: webhook from bench system; 1–2 days.
- 3. Condescending competitor-gap framing.** τ^2 -Bench cannot penalize an agent that is technically correct but socially wrong. A CTO who deliberately skipped the "missing" capability reads the gap analysis as arrogant. Probe G05 caught 60% escalation before tone-guard tuning. Fix: prospect-awareness hedge in templates; partially implemented.
- 4. Multi-thread context leakage.** τ^2 -Bench is single-threaded. Simultaneous outreach to co-founder and VP Engineering can leak context — a GDPR incident. K01/K02 confirmed correct isolation at low concurrency; untested under 50+ concurrent threads.

Public-Signal Lossiness of AI Maturity Scoring

Loud but shallow (false positive): Company posts AI thought leadership — CEO keynotes, "AI-first" letters, one "AI PM" role — and scores 2–3. Agent pitches Segment 4 (ML platform migration) to a company with no data layer. Prospect responds "we already have this." *Impact:* brand damage, wasted contact. Estimated FP rate at score ≥ 2 : **15–25%** (qualitative review of 1,001-company sample; precision/recall not yet computed against labelled data).

Quiet but sophisticated (false negative): Stealth AI startup keeps repos private, recruits by referral, scores 0. Agent sends generic email, missing the highest-margin Segment 4 engagement. **Mitigation before production:** hand-label 20–30 Tenacious past prospects to compute precision/recall.

Honest Unresolved Failure — PEV Does Not Beat Baseline

Delta A = **−0.2652** (V1 pass@1 = 0.4615 vs. baseline 0.7267; $t = -1.84$, $df = 12$, $p = 0.955$ one-sided; source: eval/score_log.json). The mechanism made performance worse. Diagnosis: the thinking model already applies chain-of-thought internally; explicit UNDERSTAND/VERIFY/PLAN instructions compete with native reasoning. V1/V2 also produced a confirmation anti-pattern (agent asked user to confirm → user said yes + stopped → tool never called → reward = 0). V3 corrected this but $n=3$ valid scores (17/20 lost to credit exhaustion) — statistically meaningless. **What to deploy:** the enrichment + policy + compose stack, not the PEV agent. The τ^2 -Bench result documents benchmark performance honestly; it does not reflect outreach pipeline readiness.

Kill-Switch Clause

KILL_SWITCH defaults **ON**; all outbound routes to staff sink until `CONVERSION_ENGINE_LIVE=true` is set by the CEO (`bash smoke_test.sh` exits 0). Pause if: (a) bench commitment not in bench summary; (b) Langfuse tone-compliance < 95% over any 7-day window; or (c) reply rate < 2% after 500 contacts.